ZAFIR MARICAR

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Skills

- C/C++ | Python | Java | Git | Azure | Cloud Computing | SQL | OpenGL | Vulkan | Docker | Backend | Anylogic | RTSP Android Studio | Assembly | VR | Socket Programming | Lua | PhysX | CUDA | AWS | Project Management | AI | Video Processing

Education

Master of Technology (Scholarship) — **National University of Singapore** — **2025 - Present**

- Major in Software Engineering (CGPA 5.0/5.0)

Bachelor of Science with Honours — **Singapore Institute of Technology** — **2021 - 2025**

- Major in Computer Science in Real-Time Interactive Simulation

Experience

AI Engineer — **NCS** — **2025 - Present**

- Architected GStreamer/FFmpeg pipelines for RTSP and file ingestion to support real-time AI inference.
- Built Docker microservices using RabbitMQ, Redis, MariaDB & Python for scalable video chunking.
- Increased concurrent video processing capacity by 30% via multi-threading and memory buffer optimizations.
- Proficient in optimizing parallel computing workloads across Nvidia CUDA and Huawei Ascend (CANN) architectures.
- Showcased live Video Analytic applications at industry conferences & experienced in deploying end-to-end AI products on-site.

Research Engineer Intern — **Hitachi Asia** — **2024 - 2025**

- Developed Proof of Concepts using Java based simulation software (Anylogic) in collaboration with the mobility team.
- Created complex transport simulation models that integrate human behavior, pedestrian flow, and transport infrastructure.
- Co-authored award-winning paper (3rd place for best paper) presented at ITS Asia-Pacific Forum 2025, Suwon, Korea.

Ground Response Force Police Officer (National Service) — **Singapore Police Force** — **2019 - 2021**

- Responded to a wide range of real world emergency incidents, saving lives and protecting property.
- Patrolled designated areas, manned neighborhood police stations and police posts independently.
- Engaged in covert law enforcement operations and served in the COVID-19 special task force.

Publications

Mo, Z., **Zafir, M.**, & Bahoy, G. L. (2025). "Influence of Selective Security Check on Heterogeneous Passengers at Metro Stations." Engineering Proceedings. [https://doi.org/10.3390/engproc2025102003]
Winner of 3rd Place Best Paper Award, presented at the 2025 Suwon ITS Asia Pacific Forum, South Korea.

Projects

Foodkaki (AI Web App)

- Lead a team of 4 as part of the AngelHack hackathon hosted by AngelHack & DBS.
- Developed an AI-powered web app using Node.js, Tailwind, Next.js, Gemini API, and Python.
- The app helps users make informed financial decisions on food purchases, leveraging generative AI

Torque Engine (3D Game Engine)

- Coded from scratch in C++, Lua & Python using OpenGL, Imgui, FMOD, Figma, Azure & Nvidia PhysX.
- Lead as Product Manager, directing engine development of a team of 7 programmers and 3 designers.
- Solely responsible for the engine's physics and networking system implementations.

IPPT Buddy (Android App)

- A fitness tracking application coded in Android Studio using Kotlin & Firebase integration.
- The application is capable of monitoring sit-ups, push-ups and running exercises in real-time.
- Took charge of sensor programming - utilizing the gyroscope, accelerometer and proximity sensors.